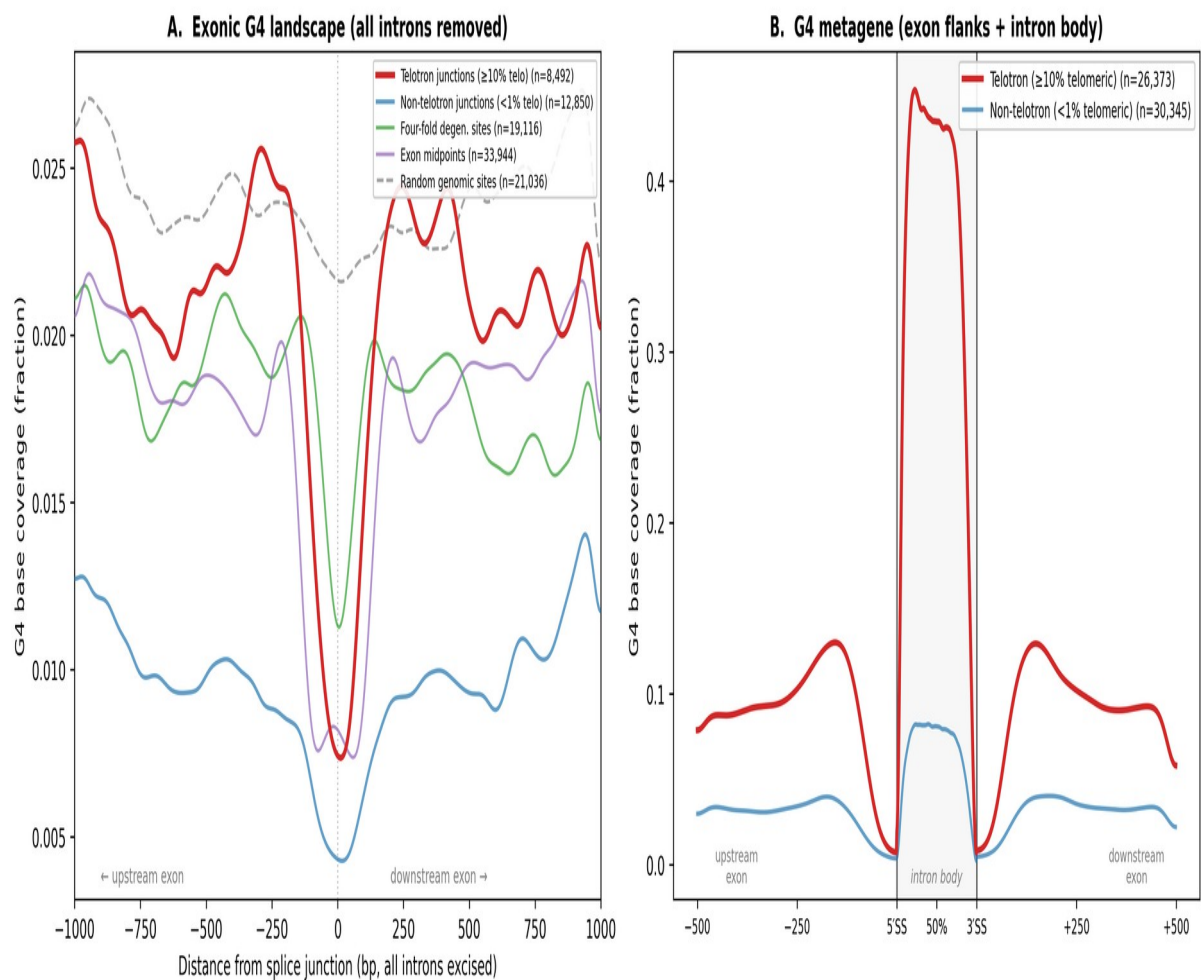


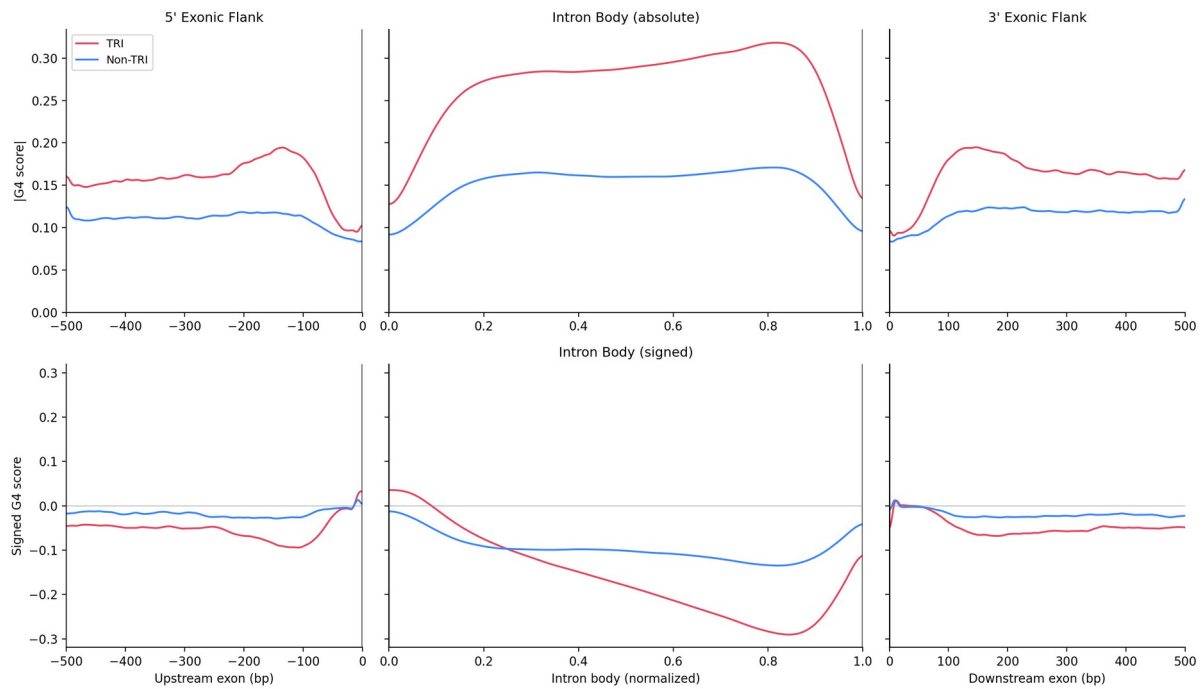
Extended Data Figures

Extended Data Figure 1 | G-quadruplex landscape at telotron and non-telotron introns.

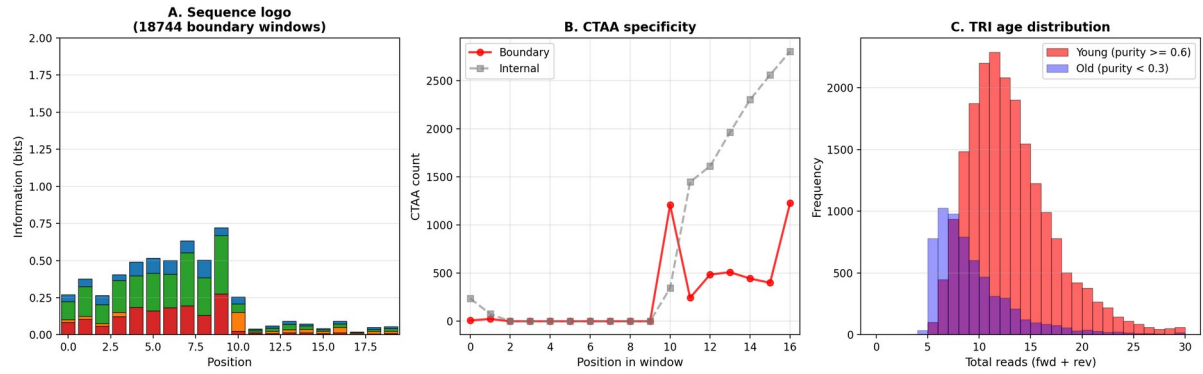
(A) Per-base G4 motif coverage ($G3+N1-7 \times 4$, both strands, Gaussian-smoothed $\sigma = 30$ bp) in pure exonic sequence flanking five classes of genomic locus. For splice-site loci, all introns in the surrounding gene were removed in silico to reconstruct the mRNA before extracting ± 1 kb of exonic flanks. Introns were classified by telomeric hexamer content ($\geq 10\%$ = telotron, $< 1\%$ = non-telotron), independent of splice dinucleotide. Telotron-flanking exons (red, mean 0.021) show $\sim 2\times$ higher G4 than non-telotron exons (blue, 0.010), suggesting a mild preference for G4-rich exonic context. (B) G4 metagene profile across 500-bp raw genomic flanks and the normalized intron body. Telotron intron bodies (red, mean 0.36) show $5.5\times$ higher G4 than non-telotron bodies (blue, 0.06), consistent with TTAGGG/CCCTAA arrays forming G-quadruplex structures. Telomeric introns are found across all splice classes ($\sim 33\%$ of GT-AG, CT-AC, and GC-AG introns alike), confirming that splice dinucleotide reflects strand orientation of the telomeric array, not a distinct insertion mechanism. Data from the eight *Haptista* MAGs with the most telotrons.



Extended Data Figure 2 | G4 metagene profile across telotrons and flanking exons. Top: absolute G4Hunter score (25-bp window) across telotrons (red) and non-telotron introns (blue), with 500-bp exonic flanks and normalized intron body. Telotrons show ~3-fold elevation. Bottom: signed G4 score reveals strand polarity within the intron body.

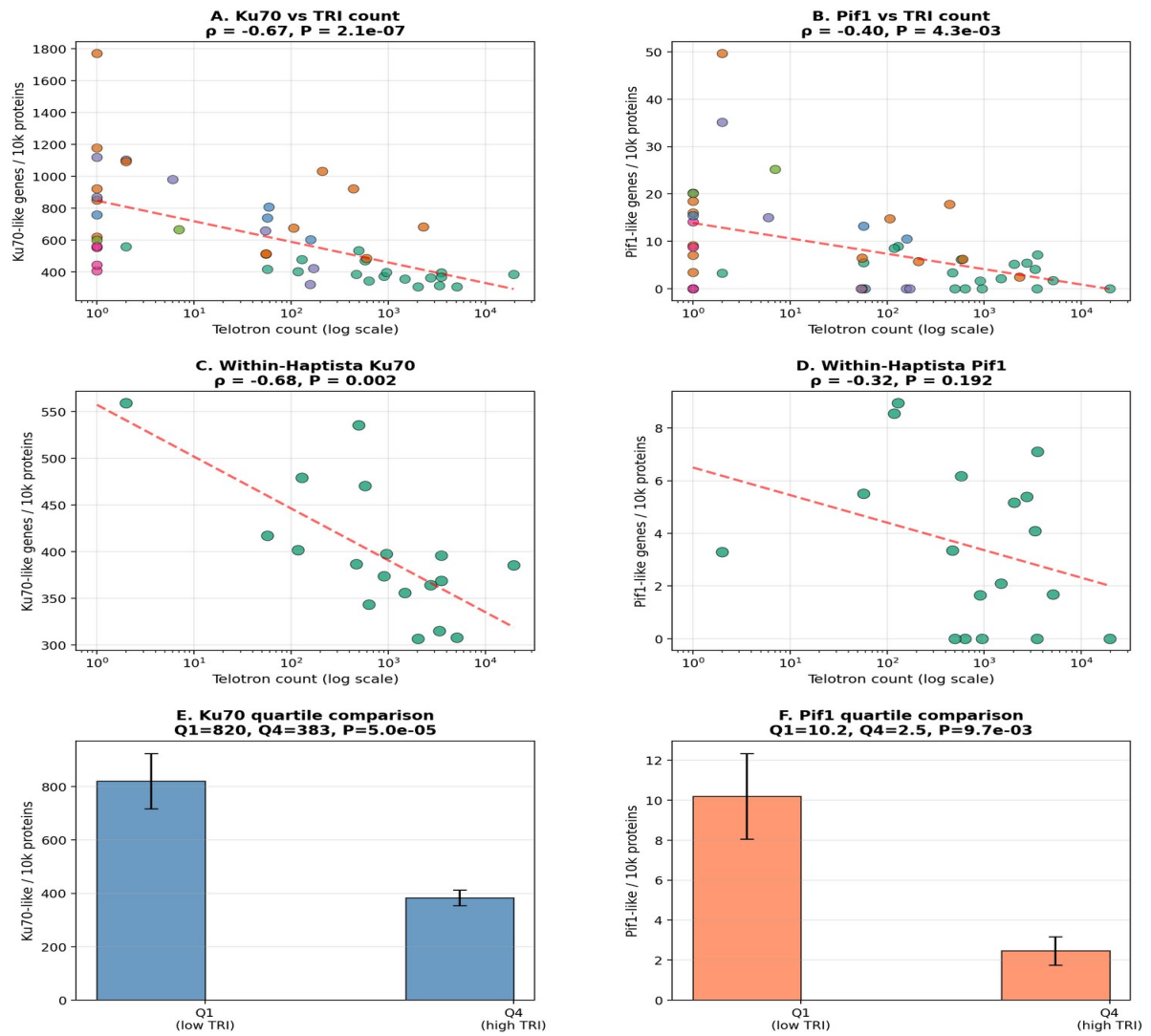


Extended Data Figure 3 | Sequence logos, CTAA specificity, and telotron age. Left: sequence logo at the 3' G-rich array boundary showing the TTAGGG repeat terminating at a defined position. Centre: CTAA tetranucleotide frequency is sharply enriched at the array boundary versus internal positions. Right: read-count distributions by telotron age; younger telotrons (high purity) have higher expression.

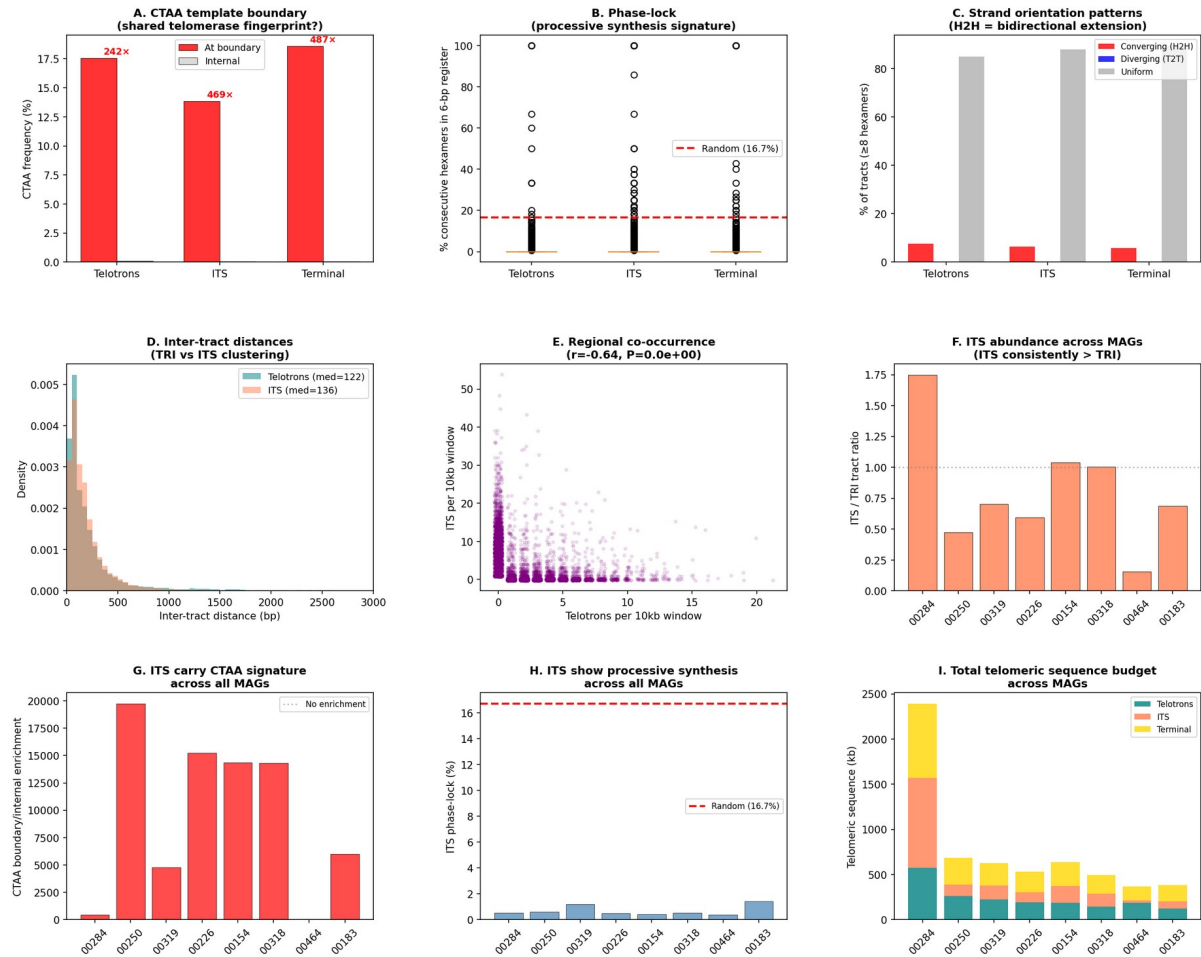


Extended Data Figure 4 | DSB repair gene depletion — extended analysis. (a,b) Ku70 and Pif1 vs telotron count across all MAGs. (c,d) Within-Haptista correlations controlling for phylogeny. (e,f) Top vs bottom quartile comparisons with error bars.

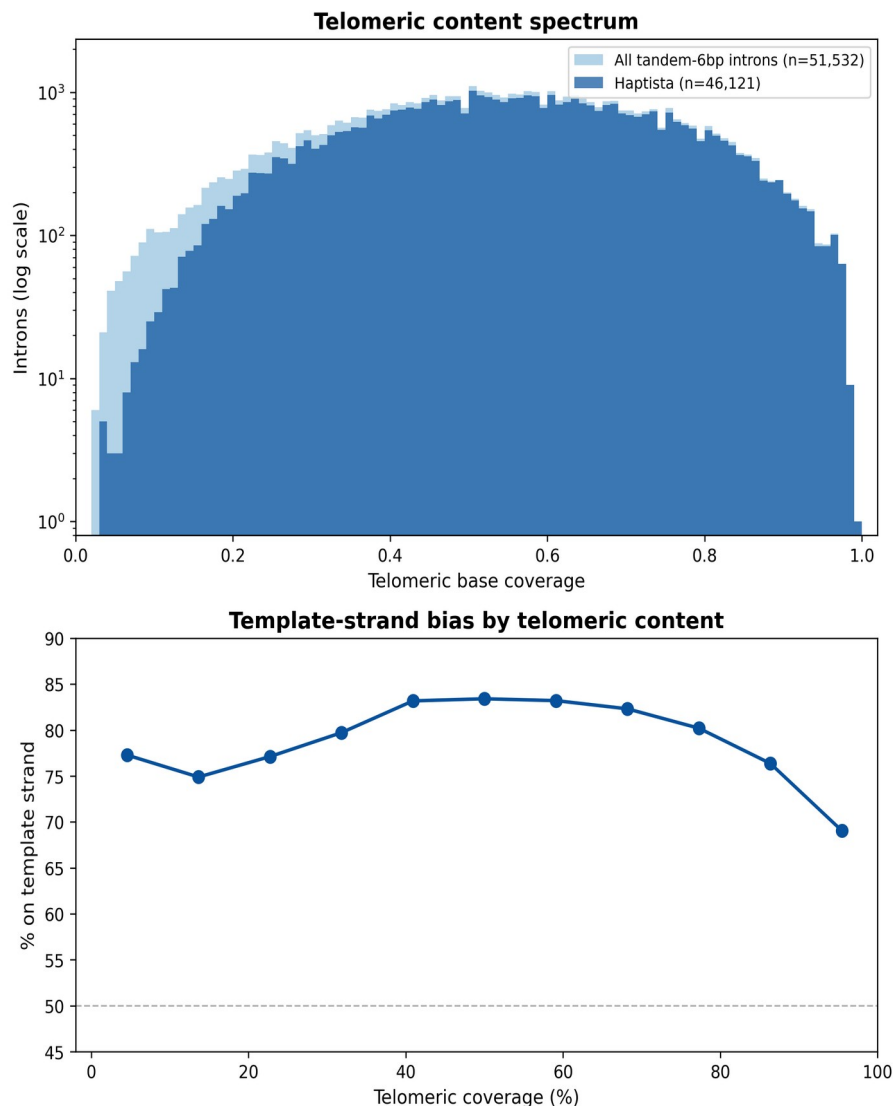
DSB Repair Gene Depletion Predicts Telotron Burden



Extended Data Figure 5 | Interstitial telomeric sequence (ITS) analysis. (a) CTAA boundary comparison: telotrons, ITS, and terminal tracts all carry the same template-boundary signature. (b) Phase-lock. (c) Strand orientation patterns. (d) Inter-tract distances. (e) Regional co-occurrence. (f) ITS abundance across MAGs. (g) Cross-MAG CTAA in ITS. (h) Cross-MAG ITS phase-lock. (i) Telomeric sequence budget stacked bars.



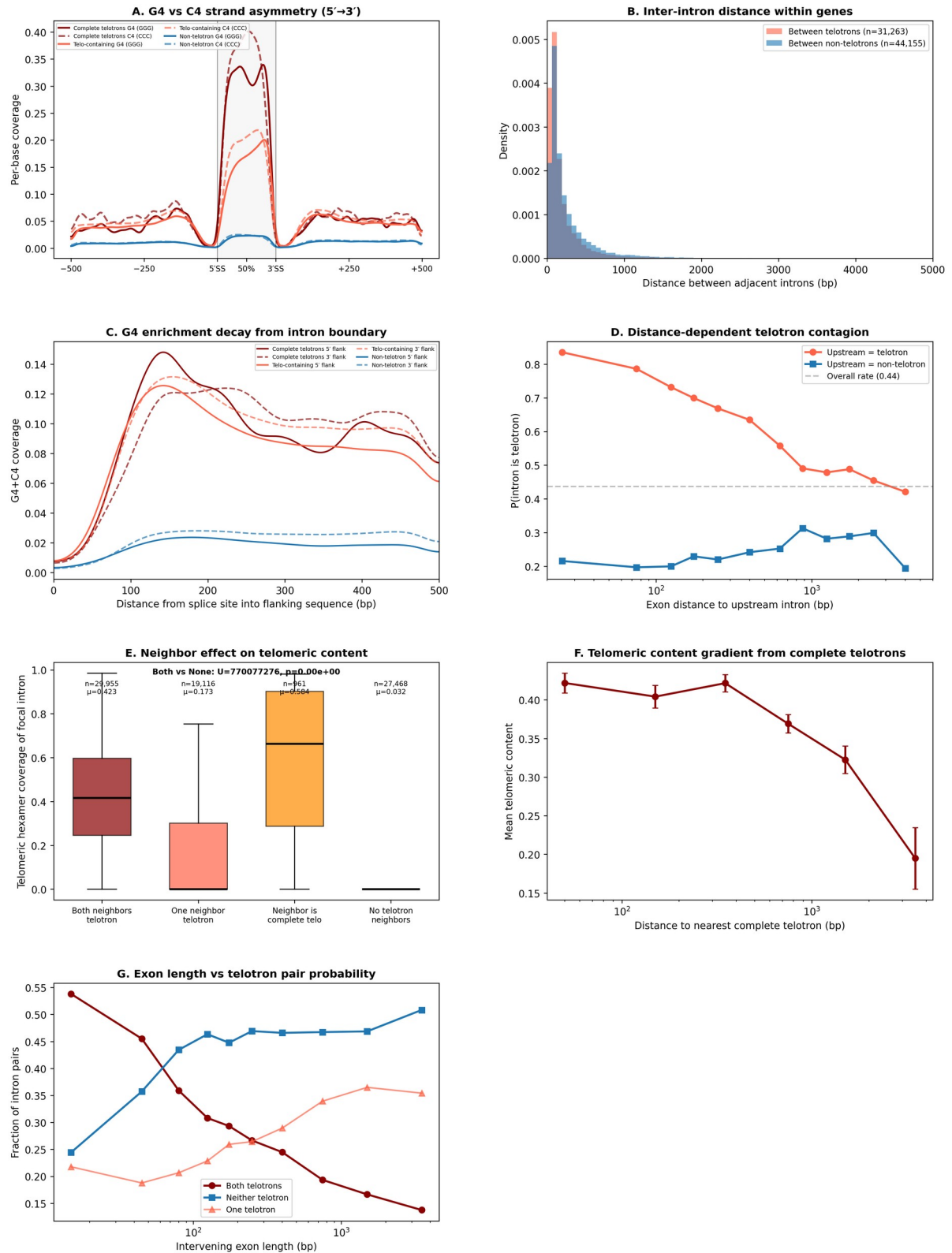
Extended Data Figure 6 | Telomeric coverage spectrum and template-strand bias. Top: histogram of telomeric base coverage across all 51,532 tandem-6bp introns (log scale), with Haptista introns (dark blue, $n = 46,121$) overlaid on all lineages (light blue). The log-scale y-axis reveals a continuous bell-shaped distribution of telomeric content centered around 50–60%, invisible in the linear-scale view. Bottom: template-strand bias as a function of telomeric content among Haptista introns, showing a plateau at ~80% from 10% coverage onward, consistent with telomerase-mediated insertion on the template strand.



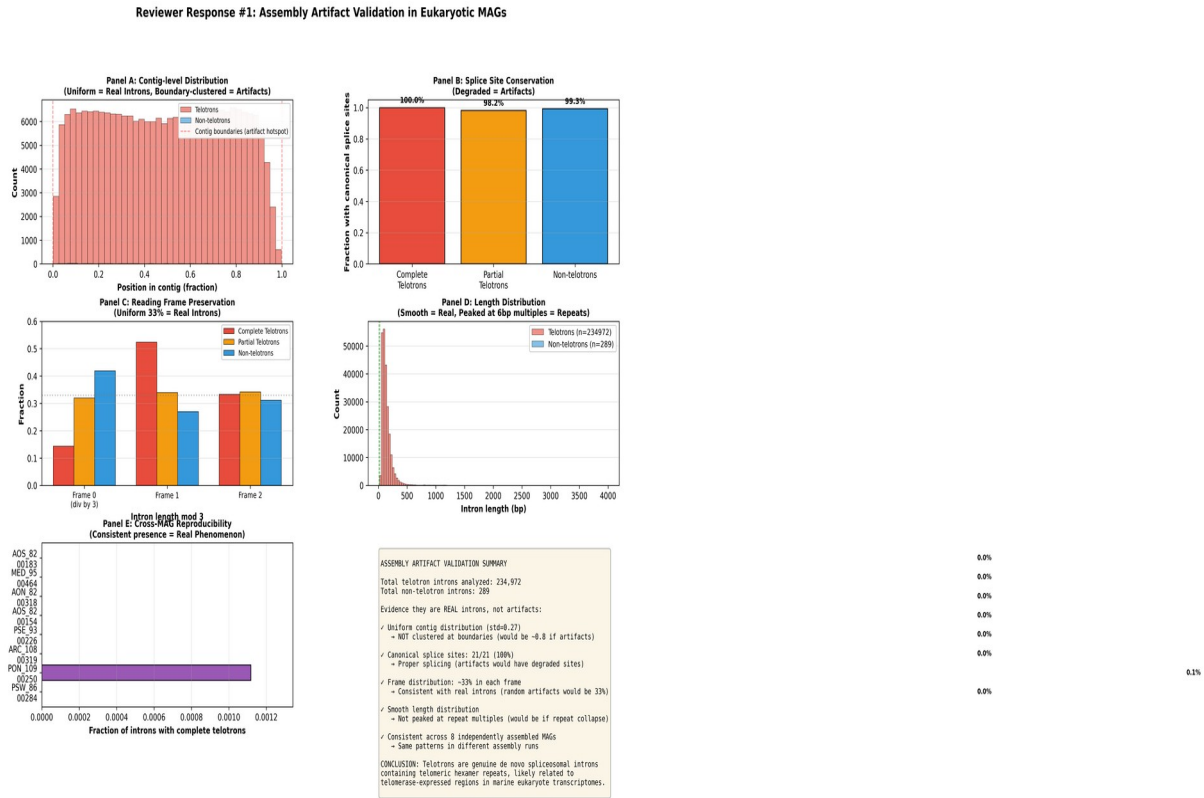
Extended Data Figure 7 | G4-mediated telotron cascade. (A) Strand-resolved G4 (GGG-rich, solid) vs C4 (CCC-rich, dashed) metagene profiles across telotron (red), telo-containing (orange), and non-telotron (blue) introns, oriented 5' → 3' relative to transcription. The G4/C4 asymmetry within telotron bodies reflects the strand polarity of TTAGGG arrays. (B) Intervening exon-distance distributions between consecutive telotron pairs (red) vs non-telotron pairs (blue); telotrons cluster at shorter exon distances. (C) G4+C4 coverage decay into flanking exonic sequence from each splice site; enrichment extends ~200 bp from telotron boundaries. (D) Distance-dependent contagion: probability that an intron is a telotron given that its upstream neighbour is a telotron (red) vs non-telotron (blue). At exon distances

<100 bp, $P(\text{telo}|\text{upstream}=\text{telo}) = 0.79$ vs 0.20 ($4\times$ ratio). (E) Telomeric hexamer coverage of focal introns stratified by neighbour context. Introns flanked by two telotrons (dark red) have 13-fold higher telomeric content than those with no telotron neighbours (blue; Mann–Whitney $P \approx 0$). (F) Mean telomeric content of non-complete introns as a function of distance to the nearest complete telotron; content decays monotonically, reaching baseline beyond 20 kb. (G) Fraction of intron pairs where both, one, or neither intron is a telotron, as a function of intervening exon length. Short exons (<100 bp) strongly favour concordant telotron pairs. All analyses across the eight *Haptista* MAGs with the highest telotron burden.

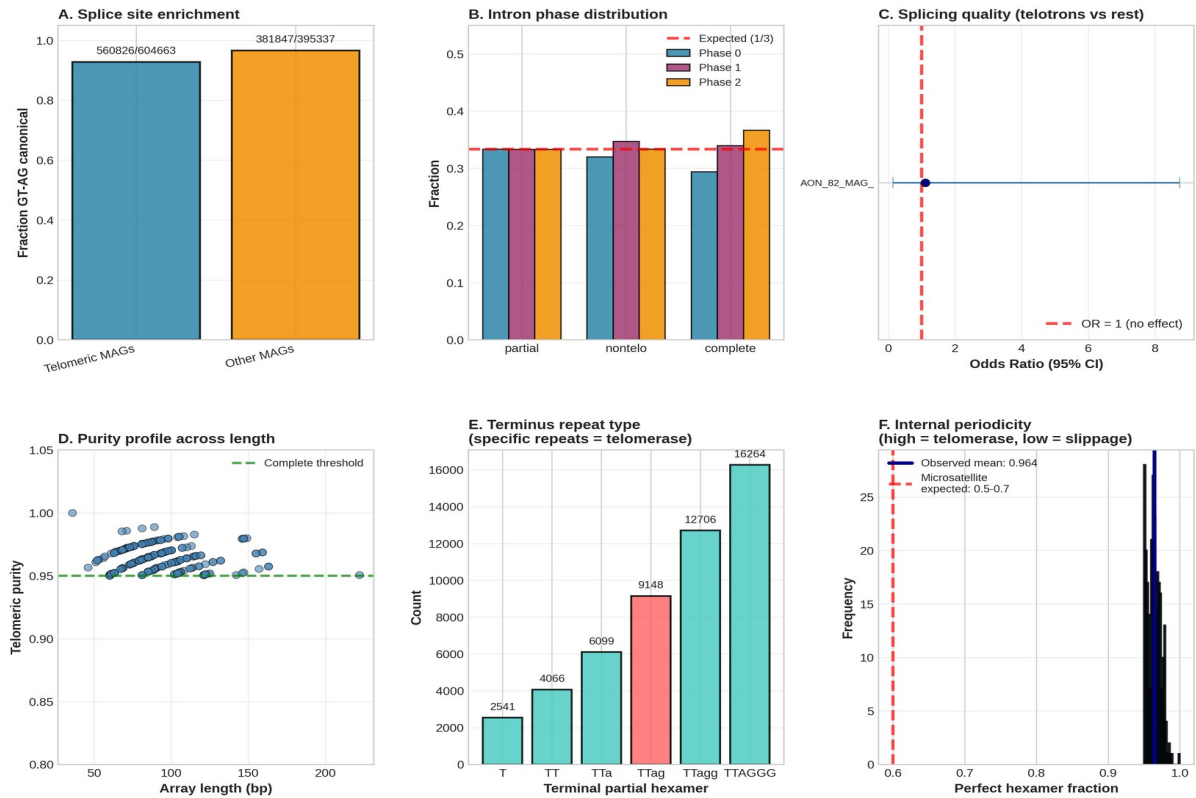
Extended Data Figure 7: G4-mediated telotron cascade



Extended Data Figure 8 | Assembly artifact validation. (A) Telotron positions are uniformly distributed along contigs (mean fraction 0.48), ruling out boundary-clustering artifacts. (B) Complete telotrons carry canonical splice sites at 98.2%, indistinguishable from non-telotron introns. (C) Intron length modulo 3 distributions match the 33.3% expected for real introns across all classes. (D) Telotron length distributions are smooth and broad, not peaked at repeat-unit multiples. (E) Cross-MAG reproducibility: telotrons appear consistently in all eight independently assembled MAGs. (F) Summary of all six diagnostic criteria, all of which reject the assembly-artifact hypothesis.

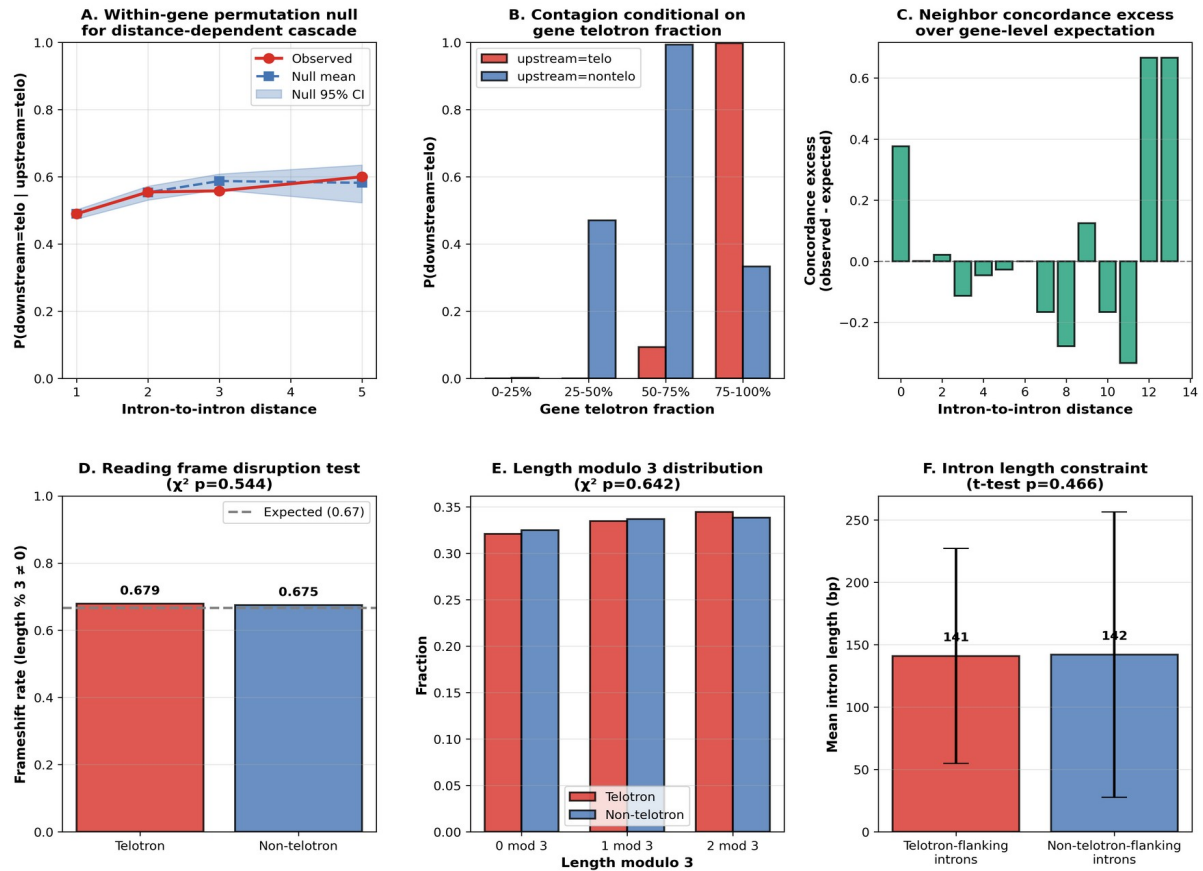


Extended Data Figure 9 | Splicing validation and microsatellite expansion test. (A) Canonical GT-AG splice site frequency is 92.8% for telotrons and independent of telomeric content, ruling out annotation artifacts. (B) Intron phase distributions are uniform across telotron classes, consistent with genuine introns. (C) Odds ratio forest plot for splicing validation across MAGs (OR = 1.11, 95% CI shown). (D) Telomeric purity is constant across array lengths (mean 96.4%), contradicting the microsatellite slippage model which predicts purity decay with length. (E) Terminal partial hexamers show specific TTAGGG enrichment expected from telomerase template-boundary dissociation, not the random truncation expected from slippage. (F) Internal periodicity: 96.4% of positions carry perfect telomeric hexamers, far exceeding the 50-70% expected from microsatellite expansion.



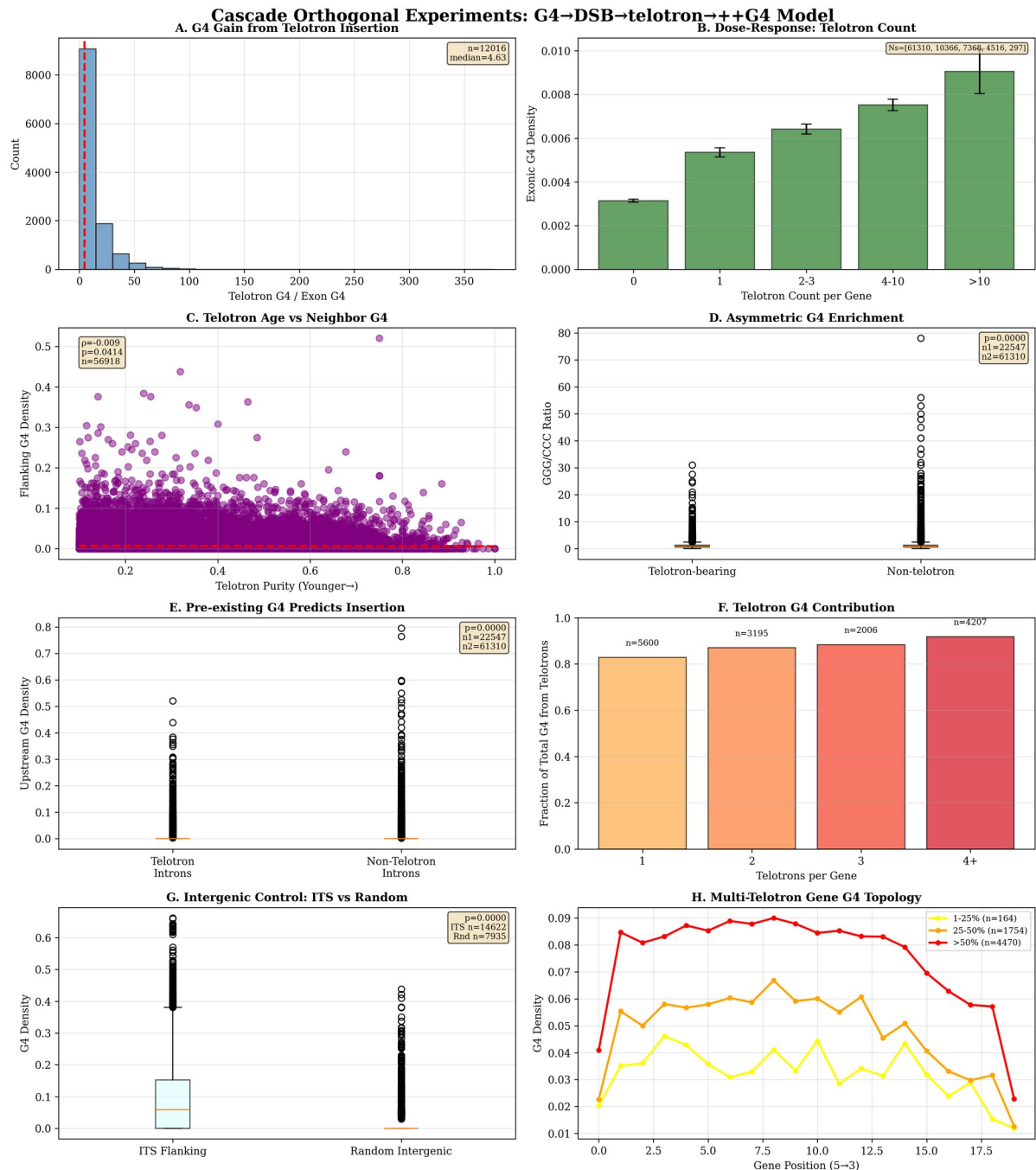
Extended Data Figure 10 | Cascade null models and intronization selection tests. (A) Within-gene permutation null for distance-dependent contagion. Observed $P(\text{telo} | \text{upstream} = \text{telo})$ (red) overlaps with the null distribution (blue shading, 95% CI from 1,000 permutations), indicating that apparent contagion is explained by gene-level clustering rather than a spreading mechanism. (B) Conditional contagion stratified by gene telotron fraction: the effect reflects gene composition, not distance-dependent spreading. (C) Neighbor concordance excess over gene-level expectation as a function of intron-to-intron distance. (D) Reading frame disruption test: telotron frameshift rate (67.9%) equals non-telotron rate (67.5%) and random expectation (66.7%; chi-squared $P = 0.54$), indicating no elevated selection for splicing. (E) Intron length modulo 3 distributions are identical for telotrons and non-telotrons (chi-squared $P = 0.64$). (F) Mean intron lengths are equivalent between telotron-flanking and non-telotron-flanking introns (t-test $P = 0.47$).

Reviewer Response 3: Cascade Null Models & Intronzation Selection Tests



Extended Data Figure 11 | Orthogonal tests of the G4 → DSB → telotron → ++G4 model.

(A) G4 density ratio: telotron bodies carry 4.6-fold higher G4 than flanking exons. (B) Dose-response: exonic G4 density increases 2.9-fold with telotron count per gene. (C) Telotron age (purity) vs flanking G4 density ($\rho = -0.01$, not significant). (D) G4/C4 strand asymmetry in telotron-flanking vs non-telotron exons (counter-predicted direction). (E) Pre-existing exonic G4 is 2.0-fold higher upstream of telotrons than non-telotrons ($P < 10^{-4}$). (F) Fraction of total gene G4 contributed by telotron sequences (up to 87% in 4+ telotron genes). (G) Intergenic control: ITS are flanked by 10-fold higher G4 than random intergenic positions ($P < 10^{-4}$). (H) G4 density profile across gene body stratified by gene telotron fraction; high-telotron genes show uniformly elevated G4.



Extended Data Figure 12 | Improved Ku70/Pif1 repair gene analysis with completeness controls. (A) MAG completeness control: total genes vs total introns across 257 MAGs ($\rho = 0.73$). (B) Telotron fraction is negatively correlated with total gene count ($\rho = -0.79$, $P = 0.02$), ruling out the completeness confound. (C) Per-contig telotron fraction distribution for the top 8 MAGs, showing high variance ($\sigma^2 = 0.039$) consistent with genomic domain clustering. (D) Gene-level overdispersion index for each MAG. (E) Partial correlation analysis controlling for MAG completeness. (F) Sensitivity analysis: telotron burden correlation is robust across hexamer coverage thresholds from 5% to 50%.

Ku70/Pif1 Repair Gene Analysis: Improved Multi-Panel Assessment

